Virtual Exploration for Blind People to Perceive Geometric and Spatial Features

Masaya Kubota
Waseda University
Tokyo, Japan
kubota_waseda@akane.waseda.jp

Masaki Kuribayashi*
Waseda University
Tokyo, Japan
Miraikan, The National Museum of Emerging Science and Innovation
Tokyo, Japan
rugbykuribayashi@toki.waseda.jp

Renato Alexandre Ribeiro*
Miraikan, The National Museum of Emerging Science and Innovation
Tokyo, Japan
renato.ribeiro@jst.go.jp

Shigeo Morishima
Waseda Research Institute for Science and Engineering
Tokyo, Japan
shigeo@waseda.jp

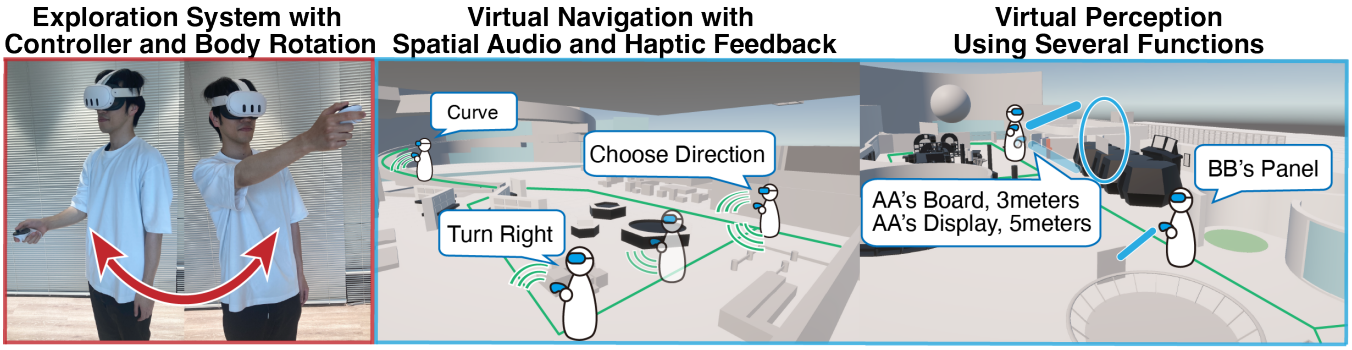


Figure 1: To investigate how route and spatial features can be perceived, we develop a VR system that supports blind people in exploring practical environments and spatial-geometric characteristics.

Abstract

Virtual reality (VR) shows promise in supporting blind people in exploring unfamiliar environments before visiting them in person through multimodal interaction. However, effective exploration requires understanding both geometric route structures and spatial features, raising questions about how blind users perceive these different forms of information and use multiple interaction methods during exploration of complex virtual environments. We present a VR exploration system to convey route structures and spatial features and report findings from two studies with blind participants. In a formative study with six participants, users explored non-linear routes; turns close to 90° and curved segments were recognized more frequently, while route reconstruction became more difficult as route complexity increased. In a second study with seven participants, users explored a virtual museum containing landmarks and objects, perceived approximately 60% of key landmarks, and actively combined multiple interaction methods according to their exploration goals and the spatial scale of information they sought. During the subsequent museum visit, participants recognized approximately half of the landmarks conveyed during virtual exploration when prompted with their names (i.e., landmark name). Across both studies, participants developed stronger knowledge of nearby objects, landmarks, and local route relationships than of how these elements related across the broader environment.

CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools**.

Keywords

Blind People, Virtual Reality, Spatial Navigation, Environmental Exploration, Accessibility

*These authors contributed equally to this work.

This work is licensed under a Creative Commons Attribution 4.0 International License.
VRST '26, Sendai, Japan
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2811-2/2026/11
https://doi.org/10.1145/3822517.3848690

ACM Reference Format:
Masaya Kubota, Masaki Kuribayashi, Renato Alexandre Ribeiro, and Shigeo Morishima. 2026. Virtual Exploration for Blind People to Perceive Geometric and Spatial Features. In *32nd ACM Symposium on Virtual Reality Software and Technology (VRST '26), November 16–18, 2026, Sendai, Japan.* ACM, New York, NY, USA, 12 pages. https://doi.org/10.1145/3822517.3848690

1 Introduction

Moving independently through an unfamiliar environment requires developing an understanding of where one can travel and what is located along the way, yet acquiring this information remains challenging for blind people [30]. Despite these challenges, blind people travel to unfamiliar locations independently [46], motivating technologies that support both preparation before a visit [22, 71] and on-site exploration support [32]. Virtual Reality (VR) offers immersive experiences through multimodal interaction and feedback, and has shown potential to support blind users in virtual *exploration* [44, 58, 64, 69], and preparation before visiting real-world environments [6, 22, 63].

Prior work in non-visual exploration suggests that understanding an unfamiliar environment involves multiple forms of spatial information. In this work, we focus on two complementary components: 1) *Geometric route structures* - capturing paths, turns and their connections. Prior systems have supported virtual route learning [13, 22, 55], but many have represented routes primarily through sequential segments and turning actions or evaluated relatively simple layouts. Real-world environments, such as public buildings, can instead contain multi-turn routes, varying turn angles, intersections, and curved paths. 2) *Spatial features* - representing landmarks and objects [4, 19, 20, 54]. Existing systems explored various approaches to support object perception [12, 14, 41, 72] with various interaction techniques in environments, such as object search (e.g., in a corridor [40, 60] or a small room [33]). Together, geometric route structures and spatial features provide complementary information for understanding an unfamiliar environment. As these components provide distinct yet complementary forms of spatial information, they raise questions about how blind users perceive each form of information and employ multiple interaction methods during exploration of complex virtual environments. Therefore, through the development of a VR exploration system to convey route structures and spatial features, we investigate how blind users perceive these environmental components and how they employ multiple interaction methods to acquire environmental information during exploration. Rather than evaluating route and feature learning simultaneously, we study these components in two stages to characterize the perceptual challenges associated with each and to identify implications for their future integration.

We first conducted a formative study with six blind participants to examine how users perceive geometric route structures, including multi-turn and non-linear routes (Fig. 1, middle panel). The findings showed that turns close to 90° and curved segments were recognized more frequently than other turn angles, while route reconstruction became more difficult as route complexity increased. These results informed the design of the second study, in which we constrained the pre-mapped routes to turns at or below 90° and added a *Perception Module* for acquiring spatial-feature information.

We then conducted a second study with seven blind participants in a virtual model of a real science museum, followed by a visit to the corresponding physical museum. The system provided four interaction methods [17, 33, 49, 67] for perceiving spatial features. Participants freely explored the virtual museum and subsequently visited the corresponding physical environment, where landmark prompts were used to examine whether information acquired during virtual exploration was recognized in the real setting. The museum visit therefore assessed cross-context landmark recognition, while the effect of VR pre-exploration on independent wayfinding performance remains to be examined. In particular, we wanted to answer the following research questions:

- RQ1: To what extent can blind users perceive geometric route structures in complex virtual environments?
- RQ2: To what extent can blind users perceive spatial features in complex virtual environments using multiple interactions?
- RQ3: How do blind users use and combine interaction methods during exploration of complex virtual environments?

Across both studies, we observed a consistent distinction between local and larger-scale spatial understanding. Participants developed stronger knowledge of local features than of how these elements related across the broader environment. In the second study, participants also actively combined interaction methods according to their exploration goals and the spatial scale of information they sought, rather than relying on a single method. These findings suggest that virtual exploration systems should support flexible information acquisition at multiple scales, and support users in connecting local observations to regions and previously explored parts.

2 Related Work

2.1 Virtual Navigation and Exploration for Blind People

Virtual navigation can support pre-exploration by allowing blind people to acquire spatial knowledge of an environment before visiting the real location [1, 2, 16, 34, 39, 62]. Other pre-exploration approaches include tactile and audio-tactile maps, models, and verbal descriptions [26, 47, 52, 68, 70, 71]. Virtual navigation can complement these pre-exploration representations by allowing users to encounter spatial and environmental information through simulated movement and interaction before a visit. Such systems have been implemented across different form factors and interaction configurations, including handheld smartphone systems and head-mounted virtual environments. Smartphone-based systems can convey route sequences, turning actions, landmarks, and POIs using widely available hardware [22, 23], while systems such as X-Road by Thevin *et al.* [69] demonstrate that spatially tracked virtual environments can also be implemented using mobile hardware. In HMD-based VR, head orientation, controller-based interaction, spatial audio, haptic feedback, and environmental geometry can be coordinated within the same 3D environment [21, 36, 42, 58]. These capabilities allow users to orient themselves and actively access information about both routes and surrounding environmental features. Prior VR systems have established methods by examining common navigation settings, such as intersection crossing and standard point-to-point navigation tasks [58, 69]. Building on this work, we investigate

how blind users perceive route structures and spatial features while exploring complex virtual environments containing multiple turns, varying turn geometries, landmarks, and non-linear paths, such as those commonly found in public buildings.

2.2 Geometric Route Structure Perception in Virtual Environments

Prior work on conveying geometric route structures in virtual environments has generally followed two approaches: Open-world navigation, where users can move freely in any direction independently of the locomotion technique, e.g., using thumbsticks [48], custom-made equipment (e.g., treadmills [24, 35] or augmented cane controllers [24, 64, 72]) and body movement via VR headset tracking [18, 58]. While this approach offers freedom to explore, understanding the overall spatial organization of larger environments can remain challenging [50]. The second approach constrains navigation to a pre-mapped route [22, 23, 36]. Although this limits navigation freedom, it provides a stable and repeatable route structure through which users can acquire route knowledge [23], including in environments with multiple turns or non-linear segments [36]. Due to its benefits, we adopt a pre-mapped route approach to convey geometric route structure. We therefore investigate to what extent this approach supports the perception of complex route structures with multiple turns and non-linear routes.

2.3 Perception of Spatial Features for Exploration

Building on prior work on conveying geometric route structures, researchers have also explored ways to support the perception of spatial features during exploration. These approaches include scene descriptions and object localization [10, 33], as well as spatial audio, haptic feedback, and object-search interactions [12, 14, 41, 72] that help blind users perceive and interact with nearby environmental features. However, understanding an unfamiliar environment also involves relating these local features to the broader spatial structure in which they are situated. For example, Jain *et al.* [30] found that, beyond route and POI information, blind users seek information about the shape of spaces and the layout of objects within them to develop a broader understanding of unfamiliar environments. Landmarks are one type of spatial feature that can support this understanding by helping users identify approximate locations and by triggering directional decisions along a route [59]. Similarly, Orientation and Mobility (O&M) research highlights the importance of landmark knowledge in supporting navigation among people with visual impairments. Taken together, prior efforts suggest that route-level understanding and awareness of spatial features may play complementary roles during exploration. Within accessible VR systems, prior work has supported both route-level understanding [22, 23] and object-level awareness [10, 33]. More recently, systems have begun to bring these forms of support together. For example, VisionPulse [42] combines free-form discovery of regions and objects with waypoint-guided navigation and audio-haptic localization of nearby targets [42]. These developments show that navigation and environmental awareness can be supported within the same virtual environment, but it remains unclear how blind users make use of and combine multiple interaction methods when their information needs shift across spatial scales, from individual objects to nearby areas and broader environmental context. We therefore investigate geometric route structure and spatial-feature perception in separate stages, with the latter focusing on how users combine multiple interaction methods according to their exploration goals and the spatial scale of information they seek.

3 Navigation Module Implementation

Our first step was to build the *navigation module*, which conveys to blind users the geometric structure of the environment, and investigate how well blind users could perceive the shapes of pre-mapped routes. To implement the system, we used the Meta Quest 3, which supports spatial audio, gesture input via controllers, and head-orientation tracking. Route guidance itself was conveyed through spatial audio, verbal cues, and controller haptics.

3.1 Virtual Environment with Pre-Mapped Route

In the initial study, the virtual environment contained only pre-mapped routes without spatial features, allowing us to focus solely on route perception. The pre-mapped route consisted of *nodes*, representing turning points, and *edges*, representing connections between nodes. Turning points could include non-perpendicular turns (e.g., 30°, 45°, or 60°), depending on the relative positions of connected nodes. Edges could represent either straight or curved path segments connecting two nodes.

3.2 Multimodal Feedback for Guiding Users Along the Pre-Mapped Path

The system guides users along pre-mapped routes while they remain seated on a freely rotating chair, physically rotate to orient themselves, and use the VR controller for virtual locomotion and interaction (Fig. 2). Users can move by pushing the controller's thumbstick forward, with the speed set based on measurements of participants' walking pace in the physical world, which was approximately 1 m/s for all participants. Movement is permitted only when both the head and the controller are oriented within 15° of the route's direction. This design was intended to encourage forward movement while allowing users to physically rotate to align themselves with the route. We also implemented additional cues to support accessibility based on prior work [3, 20, 28, 58]. If this navigation condition is not met, the system emits an error beep. A continuous footstep sound is played during virtual movement to provide additional movement feedback.

Spatial Audio. Indicates the direction and distance of travel by emitting a sound from the target direction, with volume encoding the remaining distance. On straight paths, the sound originates from the next turning point or intersection, whereas on curved paths, it is emitted from a nearer position along the curve.

Haptic Feedback. Complements audio by using two vibration patterns: 1) brief pulsed vibrations delivered once per second (pulse duration = 0.1 s, interval = 1.0 s) when the controller is pointed in a valid direction; and 2) continuous vibration when it faces a non-navigable direction [38].

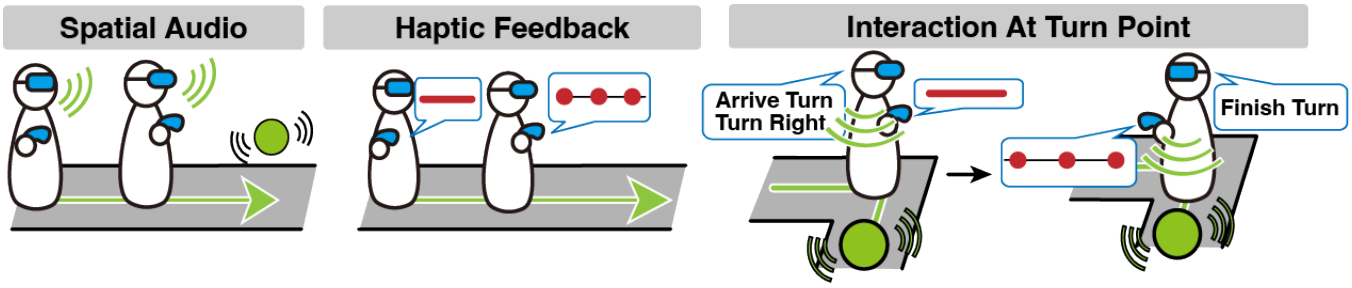


Figure 2: Navigation Module Interface and Interaction. The system uses spatial audio and haptic feedback to indicate available directions of movement. At turning points, the system informs users of the arrival and the turning direction, and the user orients their body accordingly by relying on the spatial audio and haptic feedback.

Table 1: Participants' demographics in the first user study. Light Perception = LP; Hand-Motion Perception = HMP

ID	Age	Gender	Age of Onset	Vision Status	VR Experience
P01	20	Male	14	LP in right eye	None
P02	51	Female	35	Totally blind	None
P03	24	Female	8	Totally blind	None
P04	30	Male	14	LP	None
P05	19	Male	13	Totally blind	None
P06	25	Non-binary	0	HMP	1-2 times

3.3 Interaction at Turning Points

When users arrive at a turning point, the system announces the available directions. For example, at an L-shaped corner, it states: *"You have reached a corner. Please turn right,"* while at a T-shaped intersection, it says: *"You have reached an intersection. You can go left or right."* After this initial message, the system continuously reinforces the available directions through spatial audio and haptic feedback. For example, when users point the controller and orient their head toward a valid direction, the controller vibrates and the system repeats verbal cues such as *"right"* or *"left"*. Finally, users confirm their choice by pressing the trigger button on the controller, which allows them to proceed along the selected path.

4 Navigation Module User Study

We conducted a user study to investigate how they perceive routes in a VR-based environment.

4.1 Participants

We recruited six blind participants (Table 1) through professional contacts of the research team. Participants were between 19 and 51 years old (mean=28.2; SD=11.9). All study procedures were approved by our institution's review board, and informed consent was obtained. The study lasted 180 minutes, and participants received compensation of 60 USD.

4.2 Virtual Environments

We designed eight navigation tasks, each containing a predefined ground-truth trajectory (Fig. 3). The trajectories included straight segments, turns of 30°, 45°, 60°, and 90°, curved paths, and intersection-based layouts. These trajectories were selected to represent a range of navigational challenges commonly encountered during indoor wayfinding, from simple directional changes to more complex routes involving multiple turns and intersections. The average route length was 35.6 m. The ground-truth trajectories were predefined by the researchers to provide a consistent reference for evaluating navigation performance across participants.

4.3 Tasks and Conditions

Participants first navigated a pre-mapped route using our system (**guided navigation phase**) and then reproduced the same route from memory (**reconstruction phase**). The guided navigation phase represented the primary use of our system, where participants followed a path using spatial audio, haptic feedback, and footstep sounds, as described in the previous section. In the reconstruction phase, they attempted to recall and reproduce the same path in an open virtual space, using only the thumbstick, body rotations, and footstep sounds as cues. The headset's display remained enabled throughout both studies so that the experimenter could monitor the virtual environment through screen share.

4.4 Procedure

We first conducted a 15-minute pre-interview to collect demographics, followed by a 30-minute system tutorial. Between study phases, participants were asked whether they wanted to take a short break. Then, participants completed eight tasks. The order of the routes was counterbalanced, and the topologies were mirrored relative to participants' starting orientation. For example, a right-left turn sequence could also be presented as a left-right sequence. Adopting a think-aloud protocol [65], participants were instructed to verbalize what they perceived and thought. Finally, we conducted a 40-minute open-ended post-interview.

4.5 Results and Discussion

4.5.1 Route Trajectory Analysis. To measure reconstruction accuracy, we compared reconstructed and guided trajectories using normalized Dynamic Time Warping (nDTW) [29], a metric for trajectory comparison [25, 27, 56, 57, 66]. nDTW ranges from 0 to 1, where higher values indicate greater similarity between trajectories.

Fig. 3 visualizes the reconstructed routes and deviations. Participants reconstructed routes with turns up to 90° with greater similarity to the ground truth trajectories than the routes with

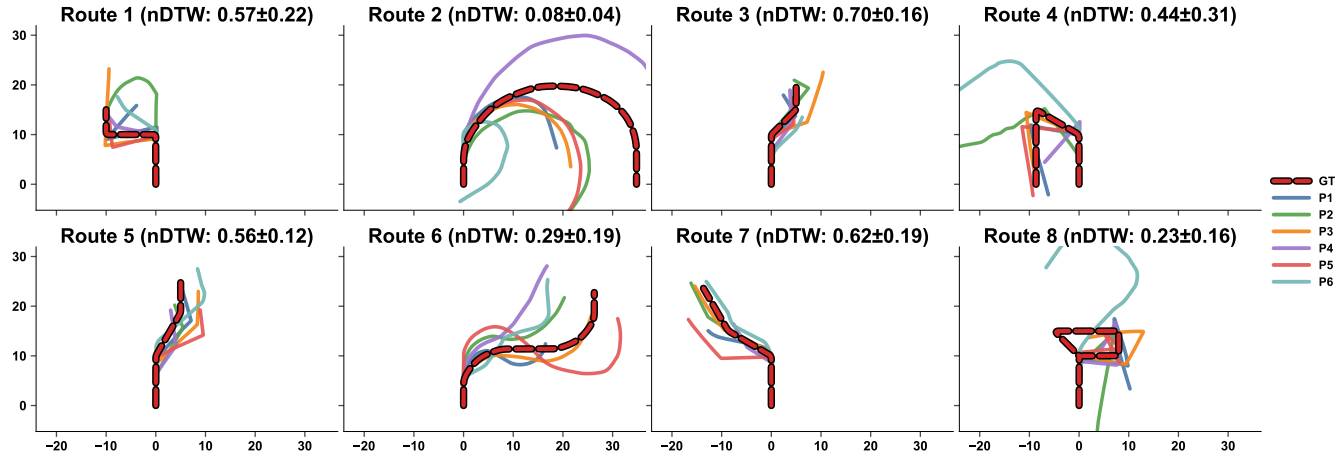


Figure 3: Participant Trajectories Compared to the Ground-Truth Path. Routes 1, 3, 5, and 7, which contain turns at or below 90°, generally showed greater similarity to the ground-truth trajectories. In contrast, Route 4, which includes a turn greater than 90°, and Route 8, which includes four turning points, showed larger reconstruction errors.

multiple turns, obtuse angles, or curves. The figure shows the nDTW variation across routes, with Route 2 having the lowest mean nDTW[1], while Route 4 has the largest variance. We did not conduct a statistical analysis given the small sample size.

4.5.2 Think-Aloud Protocol Analysis. We counted all statements referring to each turning point or curve, and calculated recognition accuracy. A turn was scored correctly if the reported angle deviated by no more than 10° from the actual value. Curves were correct if described appropriately (e.g., *"it was a semicircle"*). Recognition accuracy was 64.3% for 90° turns, 47.6% for turns less than 90°, 8.33% for turns greater than 90°, and 66.7% for curves. Participants recognized 90° turns and curves more frequently than other turn angles and could capture overall curvature. Larger turns (e.g., 120°) were often difficult to perceive, and routes with four turns were all described as three turns.

4.5.3 Effectiveness of Multimodal Feedback. Participants highlighted the effectiveness of multimodal feedback, noting that each modality conveyed distinct information. Five participants reported using spatial audio for overall orientation, while P01 relied on haptic cues for fine adjustments. **C1**:*"With spatial audio, I could perceive the general position and distance, and with the vibration, I got the finer details."* This aligns with previous non-VR literature [37, 38, 43], supporting the validity of our system design.

4.6 Findings Summary

The main findings of this study are as follows:

- Turns close to 90° were recognized more frequently than smaller or larger turn angles, while routes with multiple turns were more difficult to reconstruct.
- Although curves were not reconstructed precisely, participants recognized them.
- Spatial audio and haptic feedback played complementary roles during exploration, with spatial audio supporting general direction and distance, while vibration supported fine-grained orientation adjustments.

[1]In this paper, we report the mean and standard deviation (SD) in *mean±SD* format.

5 Perception Module Implementation

Building on the formative study, we designed the pre-mapped routes used in the second study so that turns were at or below 90°, and implemented the Perception Module to develop a full exploration system. While the Navigation Module conveys route structures, the Perception Module supports the perception of spatial features through multiple interaction methods.

5.1 Interaction Modes for Active Perception

To support spatial feature perception, we added four interaction methods based on existing interaction techniques [17, 33, 49, 67]. We selected these methods to provide access to environmental information at different spatial scales. These methods provide different forms of information about the environment, ranging from information about a single object (Point Mode), groups of nearby objects (Circle Mode), objects within a user-defined area (Shock Wave Mode), and scene-level summaries (AI Caption Mode). Users can access these methods through an on-demand selection menu and switch between them throughout exploration.

5.1.1 Point Mode. Provides information about individual objects. The system identifies the object located along the controller's pointing direction. Using Unity's raycasting functionality [67], the system detects objects along this direction. If the ray intersects an object, the system retrieves and announces the object's name. Haptic feedback conveys distance: vibration is minimal at 30 m and increases linearly as the user approaches the object.

5.1.2 Circle Mode. Provides region-based exploration, allowing users to obtain information about multiple objects within a user-defined circular area (Fig. 4) [17]. The user holds the trigger button

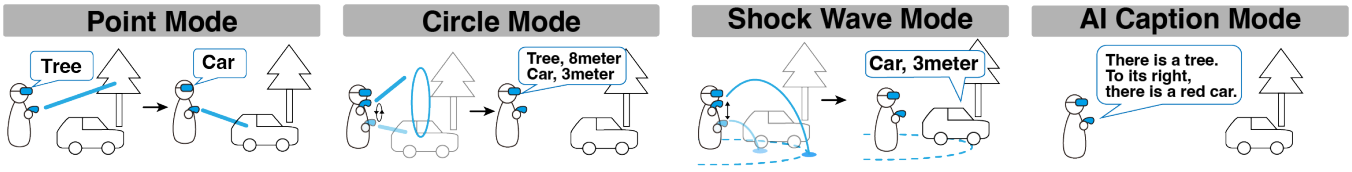


Figure 4: Interactions of the perception module.

and draws a circle with the controller. Upon release, the system identifies all objects within the drawn region within 30 meters, and their names and distances are announced.

5.1.3 AI Caption Mode. Provides scene-level descriptions of the user's current view. It uses GPT-4o [51] to describe a 1920×1080 image of the user's current field of view. While the description is generated, the system plays an audio message, *"Generating description, please wait,"* and then plays the generated message. To elicit contextually rich summaries, we engineered prompts[2].

5.1.4 Shock Wave Mode. Provides area-based exploration by announcing objects within a user-defined radius. The radius can be adjusted by tilting the controller up or down, which projects a parabolic ray onto the ground; the landing point determines the radius. The system announces the radius in 5 m increments (e.g., "5 meters, 10 meters, 15 meters"). Once confirmed with the trigger, the system retrieves all objects within the radius and reports their names and distances. Up to seven objects are conveyed using spatialized audio from each object's position, with five-second intervals and auditory cues inserted to aid comprehension.

5.2 Passive Description

In addition to active interaction, the system provides passive descriptions of predefined landmarks whenever participants pass fixed points, similar to previous navigation systems [61]. We selected key landmarks (e.g., buildings and exhibitions) and annotated them with information available from corresponding online sources.

6 Perception Module User Study

We conducted a user study to examine: 1) how blind participants perceived different environments using the provided interaction methods, and 2) the extent of information they were able to perceive. The study took place in a room at a science museum. This location was chosen because it provided a complex real environment that could be modeled for virtual exploration and subsequently revisited in the corresponding physical space.

We recruited seven new blind participants (Table 2). Five participants were totally blind, while two had limited residual vision: P07 had light perception, and P12 had hand-motion perception in the left eye and light perception in the right eye. Six had visited previously but reported limited familiarity with it, and one participant reported being familiar with the museum. None had used VR before. All sessions took approximately 150–180 minutes, and participants were compensated 60 USD.

[2]The full prompt is provided in the supplementary material.

Table 2: Participants' demographics in the second user study. Light Perception = LP; Hand-Motion Perception = HMP

ID	Age	Gender	Age of Onset	Vision Status	Museum Familiarity
P07	31	Female	23	LP	Limited Familiarity
P08	48	Female	9	Totally blind	Limited Familiarity
P09	33	Male	0	Totally blind	Limited Familiarity
P10	57	Male	53	Totally blind	Limited Familiarity
P11	50	Female	3	Totally blind	Familiar
P12	57	Male	50	Left: HMP; right: LP	Limited Familiarity
P13	21	Male	3	Totally blind	Limited Familiarity

6.1 Virtual Environment

We prepared a virtual environment of a real science museum, where the study was conducted. The environment was modeled with 3D visual and audio features and included two floors and the circular slope connecting them. We manually designed pre-mapped routes (green lines in Fig. 5), ensuring that turns did not exceed 90°, following findings from the formative study. To enhance immersion, we placed virtual sound sources throughout the environment (e.g., ambient sound and sounds produced by interactive exhibits).

6.2 Procedure

We first conducted a 15-minute pre-interview to collect demographics. We then conducted a 55-minute training session to familiarize participants with the system using a learning environment based on a publicly available city asset [15], using a pre-mapped route. The session included a tutorial on each function and a 15-minute practical exploration. Between study phases, participants were asked whether they wanted to take a short break. Throughout the virtual training and museum exploration, participants remained seated on a freely rotating chair and used the controller for virtual locomotion, as in the first study. For the main task, they were asked to freely explore the museum environment for 30 minutes with the system. Then, participants visited the actual museum they had previously explored virtually. Together with the experimenter, they visited the two floors for 15 minutes. For safety, the experimenter accompanied participants throughout the museum visit, with participants holding the experimenter's arm while walking through the museum. For each landmark that had been presented to the participant during the virtual exploration, the experimenter asked participants whether they remembered the landmark at that location by reading its name aloud (e.g., "Did you notice *[name of the landmark]* in the virtual environment?"). Participants responded based on what they had perceived during the virtual exploration. Landmarks that had not been presented were not queried.

Finally, we conducted a 40-minute post-interview consisting of both Likert-scale questions and open-ended questions, the System

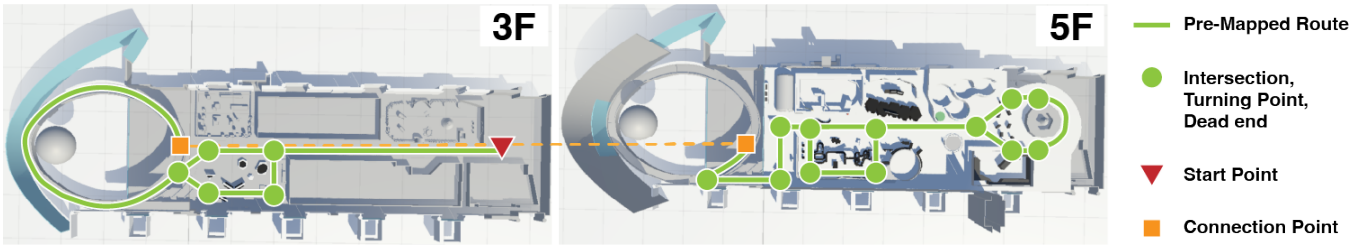


Figure 5: Environment Used in the Main Study. The environment was modeled after an actual public building, introducing complex layouts with an elevated circular route and multiple exhibitions that emit various sounds.

Table 3: Function usage distribution. Participants showed different usage patterns: P09 and P12 did not use Shock Wave Mode; P08, P09, and P10 used Point Mode less often and mainly used Circle or AI Caption Mode; P12 used Point Mode more often; and P07 used all modes in a balanced manner.

Function	P07	P08	P09	P10	P11	P12	P13	Mean±SD
Point Mode	8	1	1	0	6	13	8	5.29±4.46
Circle Mode	10	9	6	7	6	2	4	6.29±2.55
AI Caption Mode	6	7	3	10	4	5	7	6.00±2.14
Shock Wave Mode	5	7	0	2	2	0	3	2.71±2.37

Table 4: The percentage of important landmarks (e.g., exhibitions and buildings) and objects that participants perceived through the system's verbal descriptions when using the interaction modes. On average, participants perceived approximately 60.20±22.54% of the key landmarks and 26.83±10.85% of objects.

	P07	P08	P09	P10	P11	P12	P13
Objects	18.69%	36.45%	12.15%	13.08%	33.64%	34.58%	39.25%
Landmarks	35.71%	64.29%	42.86%	28.57%	85.71%	85.71%	78.57%

Usability Scale (SUS) [8], the Raw Task Load Index (Raw TLX) [9]. Because the system required participants to navigate, perceive environmental information, and switch between multiple interaction methods, SUS and Raw TLX were collected to contextualize participants' exploration experiences in terms of perceived usability and cognitive demand.

6.3 Data Analysis

Interaction logs and questionnaire responses were summarized using descriptive statistics, including frequencies, percentages, means, and standard deviations. Interviews were transcribed and analyzed using Codebook Thematic Analysis as described by Braun & Clarke [7]. Coding was iteratively refined through repeated transcript review and team discussions.

7 Results and Discussion

7.1 Quantitative Analysis

7.1.1 Function Usage Frequency. Table 3 summarizes the frequency of interaction methods used during exploration. Together with the qualitative findings, these usage patterns suggest that participants acquired information at multiple spatial scales and actively switched between interaction methods during exploration.

Table 5: Number of landmarks recognized by participants in the real museum ("Recognized" row), number of landmarks conveyed by the system during virtual exploration ("Conveyed" row; out of 14 total landmarks), and their *landmark recognition rate*, defined as the proportion of landmarks presented by the system in the virtual environment that participants later reported recognizing when prompted with the landmark name during the museum visit. The mean recognition rate was 49.86±19.82%.

	P07	P08	P09	P10	P11	P12	P13
Recognized	4	2	2	3	5	5	6
Conveyed	5	9	6	4	12	12	11
Recognition Rate	80%	22%	33%	75%	42%	42%	55%

7.1.2 Object Perception Analysis. We measured the percentage of important landmarks (e.g., exhibitions and buildings) and general environmental objects (e.g., bins and cars) that were verbally presented to participants when they used the interaction methods (Table 4). The science museum environment had 14 exhibits that served as landmarks, and 107 general environmental objects (e.g., benches, exhibit panels, and screens). On average, participants perceived 60% of the important landmarks. Although they perceived only 26% of the general environmental objects, these objects were numerous and often less relevant for exploration.

7.1.3 Landmark Recognition in the Physical Museum. Table 5 summarizes the number of landmarks recognized during the museum visit and the landmark recognition rate (i.e., the proportion of system-presented landmarks that participants reported recognizing when prompted with the landmark name). Overall, the average landmark recognition rate was 49.86%. Participants such as P07 and P10, who perceived only a small number of landmarks through the interaction modes, showed high landmark recognition rates. In contrast, P11 and P12, who were exposed to many landmarks, exhibited lower landmark recognition rates. This suggests the difficulty of retaining a large set of landmarks within a single exploration session and indicates a need for system features that support memory retention.

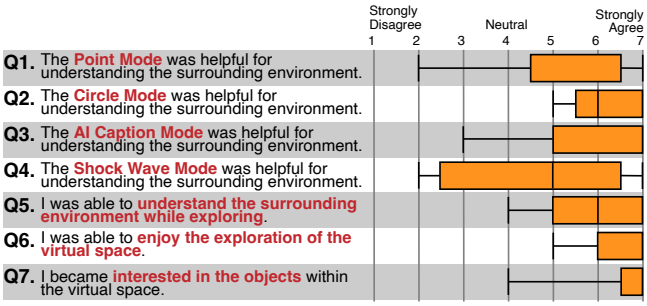


Figure 6: Levels of Understanding, Enjoyment, and Interest in the Environment Using Our System, Measured Through Questions and Seven-Point Likert Scale Responses.

Table 6: SUS and Raw TLX Scores. The Raw TLX score was computed using reverse-coded Performance values to align the directionality of all subscales.

	P07	P08	P09	P10	P11	P12	P13	Mean±SD
Mental Demand	8	6	10	8	6	3	3	6.29±2.43
Physical Demand	2	6	5	5	6	7	7	5.43±1.59
Temporal Demand	3	4	5	9	3	2	2	4.00±2.27
Performance	5	6	5	2	4	7	10	5.57±2.32
Effort	6	6	10	6	6	3	4	5.86±2.03
Frustration Level	3	5	8	1	4	6	2	4.14±2.23
RAW TLX	28	32	44	38	32	25	19	31.14±7.64
SUS	50.0	60.0	70.0	15.0	70.0	87.5	95.0	63.93±24.49

7.1.4 Likert Questionnaires. Fig. 6 Q1–4 reports the perceived usefulness of each function. Seven participants rated Point, Circle, and AI Caption Mode positively, giving scores of five or higher. The Shock Wave Mode showed the most variance, with half rating neutral or negative. As shown in Q5–7, most participants reported high satisfaction, with scores of five or higher.

7.1.5 SUS and Raw TLX Scores. Table 6 reports System Usability Scale (SUS) [8] and Raw TLX scores [9]. The average SUS was 63.93 (SD=24.49). Participants with the highest SUS, P12 (87.5) and P13 (95.0), both reported relatively low Mental Demand scores of 3. In contrast, P07 and P10, who reported lower SUS of 50.0 and 15.0, respectively, both reported Mental Demand scores of 8.

7.2 Qualitative Findings

Representative participant comments are labeled C1, C2, and so forth for reference in the discussion.

7.2.1 Combining Interaction Methods During Exploration. All participants described combining multiple interaction methods rather than relying on a single one. Participants selected and switched between methods according to the scale of information they sought, their current exploratory goals, and the amount of information already available to them. The most common strategy, reported by five participants (P07, P09, P11, P12, and P13), involved first obtaining an overview of the environment and then progressively narrowing their attention to specific objects or locations. Participants often began with methods that provided broad contextual information and subsequently used more targeted methods to confirm details or investigate areas of interest, for example: **C2**: *"First, I used Shock Wave Mode to grasp the overall environment. Then I used Circle Mode and AI Caption Mode to understand the surrounding area. When I wanted more specific information, I used Point Mode and object search to locate individual objects."* (P07)

Four participants (P08, P11, P12, and P13) also selected interaction methods according to the spatial scale of information they wanted to obtain. Nearby objects were often explored using Circle Mode or Point Mode, whereas Shock Wave Mode and AI Caption Mode were used to obtain information about more distant areas or to gain a broader understanding of the environment. **C3**: *"For nearby things, I used Circle Mode. For distant things, I used Shock Wave Mode. Shock Wave allowed me to specify larger distances, so I used it when I wanted information about objects farther away."* (P08)

P12 further adapted his strategy according to the exploration context. P12 explained that he relied more heavily on broader explanatory modes in the learning environment, where the layout was less predictable, but used more targeted methods around exhibits because he expected them to be arranged in a regular manner.

Finally, two participants (P09 and P10) explicitly described switching to another method when their initial approach did not provide sufficient information. Rather than treating individual methods as independent tools, participants moved between methods to change the scope or type of information available to them: **C4**: *"When I couldn't quite understand things using Circle Mode, I used the broader AI Caption Mode, and sometimes I did the opposite as well."* (P09)

7.2.2 Preferences for Information Delivery. Participants expressed different preferences regarding how environmental information should be delivered, while emphasizing the need to balance proactive support, comprehensive information, and user control.

For some participants (P11 and P12), environmental information was most useful when presented hierarchically, with a high-level overview provided first and more detailed information made available on demand: **C5**: *"I would like the system to first tell me what is there and then explain the details. First let me know what exists, and then provide information such as colors and layouts."* (P11)

More than half of the participants (P07, P09, P10, and P11) preferred information to be delivered proactively while moving through the environment: **C6**: *"I'd like something that automatically tells me where I am, what is on my right and left, and what lies ahead"* (P07).

At the same time, participants emphasized the importance of retaining control over information delivery. Participants (P07, P08, P09, and P10) wanted the ability to adjust or disable information when it was unnecessary and requested mechanisms for replaying previously delivered guidance: **C7**: *"It would be useful to have something like a repeat button because sometimes I missed part of the guidance while operating the system."* (P09)

Interestingly, three of these participants (P08, P09, P10) did not view proactive information delivery and user control as competing requirements. They preferred automatic information delivery that reduced interaction demands while preserving control over when and how information was presented. They described wanting descriptions of nearby surroundings to be announced without requiring frequent manual operations while still being able to customize, simplify, replay, or temporarily disable guidance when desired.

7.2.3 Virtual Exploration and Real-World Understanding. Participants reported that the information about nearby objects, landmarks, and surrounding facilities contributed to the development of mental representations of the environment. They noted that combining information from multiple system methods allowed them to construct an image of the portions of the environment they explored. **C8**:*"Because I could detect objects using various functions, I was able to build a mental image of the areas that I explored."* (P07).

P07, P08, and P11 described having a better understanding of local object-distance relationships than larger-scale spatial structures: **C9**:*"I understood what was there and roughly how far away it was, but the relationships between exhibits were difficult."* (P08)

The constrained structure of the museum environment further appeared to support spatial understanding. Both P07 and P12 remarked that corridors, walls, and restricted movement made it easier to understand their orientation and current location than in the outdoor environment. **C10**: *"As I moved through the environment-walking straight and turning corners-I gradually began to get a sense of what it was like."* (P07) During the physical visit, several participants described comparing their understanding from virtual exploration with the corresponding physical environment.

At the same time, five participants (P07, P08, P09, P10, and P12) identified discrepancies between their understanding of the virtual environment and the actual physical space. Differences in perceived distance, orientation, and route structure occasionally resulted in mismatches between participants' mental models and the real environment. Despite these discrepancies, participants described the approach as potentially useful for familiarizing themselves with unfamiliar environments before visiting them in person.

8 Discussion

8.1 Information Acquisition Strategies During Exploration

A key observation is that participants rarely relied on a single interaction method during exploration (Table 3). Prior work on accessible virtual environments has investigated route-level understanding [22, 23] and object-level awareness [10, 33] separately, and interaction methods are often evaluated as independent techniques. In contrast, participants in our study actively combined multiple methods according to their exploration goals, the spatial scale of information they sought, and the amount of information already available to them. These include obtaining an overview of the environment before narrowing their attention to specific objects (**C2**), while others moved between methods to verify information obtained through a different source (**C4**).

These findings suggest that participants did not treat individual methods as competing alternatives but rather as complementary sources of information that could be integrated to progressively construct environmental understanding. This behaviour coincides with the notion of perceptual agency [5], where understanding emerges through active engagement with the environment rather than passive reception of information. While our study was not designed to evaluate the metaphor directly, the observed exploration behaviours highlight the importance of supporting multiple complementary methods for acquiring information.

Interestingly, one participant adapted his interaction strategies according to the exploration context, relying more on broad explanatory modes in the learning environment (city) and more targeted methods around exhibits. For this participant, the less structured city environment appeared to encourage broader contextual exploration, whereas more predictable environments may encourage targeted exploration of specific features.

Participants also expressed preferences regarding how environmental information should be delivered. Some preferred information to be presented hierarchically, beginning with a high-level overview and allowing additional details to be accessed on demand (**C5**). Others desired proactive descriptions of nearby objects and route conditions (**C6**) while simultaneously emphasizing the importance of replay, filtering, and customization mechanisms (**C7**). Prior work has primarily focused on increasing access to environmental information through richer descriptions and additional interaction techniques [10, 33]. Rather than requesting additional environmental information, participants in our study primarily emphasized the ability to regulate when, how much, and what type of information was delivered (**C6**-**C7**). The challenge was therefore not access to information, but control over it. The seated setup reduced the physical demands associated with locomotion during virtual exploration, although prolonged use may still involve repeated arm and wrist movements from pointing and drawing. Longer-term studies could examine both the comfort of these interactions and whether experienced users converge on a smaller set of preferred methods. Supporting exploration may require systems that allow users to dynamically regulate the timing, amount, and scope of information according to changing exploration goals. At the platform level, one practical advantage of HMD-based VR is that current off-the-shelf headsets integrate head tracking, controller input, spatial audio, haptic feedback, and 3D rendering within a single platform. Although the feedback provided in our studies was primarily non-visual, the visual display may also provide a complementary modality for users with residual vision [73, 74]. At the same time, several of the interaction principles explored here may also inform future mobile or other wearable systems with comparable sensing and feedback capabilities.

8.2 Supporting Mental Models Beyond Local Perception

Across both studies, participants demonstrated a strong ability to perceive local features. In the formative study, participants recognized turns close to 90°, identified the presence of curves, and reconstructed simpler route structures. In the main study, participants identified approximately 60% of landmarks and reported understanding nearby objects and object-distance relationships. In the museum visit, participants recognized approximately half of the landmarks that had been conveyed during virtual exploration.

Despite investigating route structures and spatial features separately, both studies revealed a common pattern: participants developed stronger knowledge of local environmental relationships while experiencing difficulty constructing larger-scale representations. Participants often understood what objects were nearby and their approximate locations, but they found it difficult to understand how landmarks and routes related to one another across the broader

environment (**C9**). This pattern was reflected in both the qualitative findings and the route reconstruction results, where performance deteriorated as route complexity increased and turns greater than 90° were frequently misidentified. Future work could investigate additional guidance or feedback for routes with turns greater than 90° and more complex multi-turn structures (e.g., routes with several consecutive turns, intersections, and curved segments), rather than limiting route design to turns at or below 90°.

Consistent with prior mobility research [53], participants reported that the constrained structure of the environment facilitated orientation and self-localization. Corridors, walls, and restricted movement appeared to provide useful spatial anchors that were less available in the outdoor learning environment.

Interestingly, discrepancies between the virtual and physical environments were not evenly distributed across aspects of spatial understanding. Participants frequently recognized landmarks and route characteristics encountered in VR, yet reported inaccuracies in perceived scale. This pattern suggests that environmental features were more consistently recognized across the two settings, while participants more often reported discrepancies in perceived distance and scale. Distance and scale may have been harder to estimate due to artificial locomotion (i.e., using a thumbstick) and the absence of physical walking cues, consistent with prior work highlighting the importance of vestibular and proprioceptive self-motion cues during navigation and spatial learning [50, 58].

The challenge was relating environmental features to a broader structure. Route reconstructions became less accurate as the number of turns increased, and turns greater than 90° were frequently misidentified. Participants' comments indicated the same pattern. Local understanding was generally stronger, whereas broader cognitive maps remained incomplete.

These findings suggest that virtual exploration supported participants' acquisition of local environmental knowledge and familiarization with important landmarks, while larger-scale spatial representations remained more difficult to develop. This pattern is consistent with prior research suggesting that local landmark and route knowledge do not necessarily lead to accurate global representations of larger environments [11, 45]. One possible explanation is that the interaction methods primarily exposed information at a local scale, encouraging participants to incrementally construct knowledge about nearby objects and route segments rather than relationships across the entire environment.

Supporting exploration in complex environments may therefore require stronger integration between route structures and environmental features. Landmarks could be explicitly connected to travelled paths, previously explored regions, and upcoming destinations. Systems might also support awareness of explored and unexplored portions of an environment, enabling users to progressively organize local observations into larger spatial representations.

8.3 Limitations and Future Work

Our findings should be interpreted in the context of some limitations. For system development, we designed the system based on prior work and existing practices, and these choices may have shaped participants' exploration experiences, as blind participants were not involved in the design phase. Future work should engage blind collaborators earlier, for example through longer-term community partnerships [31]. Because we evaluated exploration in a single museum environment, investigating a wider range of environments could provide a deeper understanding of how users adapt their exploration strategies across different settings. Because the museum visit prompted only landmarks that had been presented during virtual exploration, the protocol did not assess false-positive recognition of landmarks and did not assess independent wayfinding. Future work should examine whether virtual pre-exploration supports independent navigation in the corresponding physical environment. The city environment was used for training rather than as a second test environment, so differences observed between the city and museum should be interpreted as individual observations rather than systematic environment-level effects. The practical role of pre-exploration may also differ across use contexts, such as preparing at home before travel, refreshing a route shortly before departure, or reviewing an environment immediately before entry. Future work could examine when blind users would choose to use such systems and what information they seek in each context.

In addition, although we examined route structures and spatial features separately, we did not investigate how users simultaneously learn these components and the relationships between them. Understanding how route and landmark knowledge are jointly acquired in complex environments may be important for supporting transfer from virtual exploration to real-world navigation. Moreover, we did not assess whether participants with residual vision made use of the headset's visual display during exploration.

Finally, participants were free to switch between interaction methods throughout exploration, reflecting realistic usage but making it difficult to determine the contribution of individual methods to environmental understanding. Future work could therefore examine how specific interaction methods, and combinations of methods, support different exploration tasks and environmental learning.

9 Conclusion

VR shows promise as a tool for supporting blind people in exploring unfamiliar environments before a real visit. We investigated two forms of spatial information relevant to exploration: geometric route structures and spatial features. Through two studies with blind participants, we examined the perception of these components separately and explored how multiple interaction methods support environmental understanding in complex virtual environments.

Findings show that participants perceived simpler route structures and recognized curved segments, while also acquiring information about important landmarks and local features. Participants actively combined multiple interaction methods according to their exploration goals and the spatial scale of information they sought. Across both studies, participants developed local knowledge but struggled to construct larger-scale spatial representations.

Findings suggest that future virtual exploration systems should support flexible information acquisition and better help users understand how nearby landmarks and routes fit within the larger environment. Our work highlights the potential of VR for supporting non-visual exploration and environmental understanding.

Acknowledgments

This work was supported by JSPS KAKENHI (24H00742, 24H00748).

References

[1] Amandine Afonso-Jaco and Brian F. G. Katz. 2022. Spatial Knowledge Via Auditory Information for Blind Individuals: Spatial Cognition Studies and the Use of Audio-VR. *Sensors* 22, 13 (2022).

[2] Kevin Allain, Bas Dado, Mick Van Gelderen, Olivier Hokke, Miguel Oliveira, Rafael Bidarra, Nikolay D Gaubitch, Richard C Hendriks, and Ben Kybartas. 2015. An Audio Game for Training Navigation Skills of Blind Children. In *SIVE*. IEEE, Piscataway, NJ, USA, 1–4.

[3] Ronny Andrade, Steven Baker, Jenny Waycott, and Frank Vetere. 2022. A Participatory Design Approach to Creating Echolocation-Enabled Virtual Environments. *TACCESS* 15, 3 (2022).

[4] Dina Bakir, Yasser Mansour, Shaimaa Kamel, Yasser Moustafa, and Marwa H Khalil. 2022. The Spatial Experience of Visually Impaired and Blind: An Approach to Understanding the Importance of Multisensory Perception. *Civil Engineering and Architecture* 10, 2 (2022), 644–658.

[5] Harshadha Balasubramanian, Cecily Morrison, Martin Grayson, Zhanat Makhataeva, Rita Faia Marques, Thomas Gable, Dalya Perez, and Edward Cutrell. 2023. Enable Blind Users' Experience in 3D Virtual Environments: The Scene Weaver Prototype. In *CHI'EA*. ACM, New York, NY, USA.

[6] Katrina Blake, Helen Kinnane, and Melinda Whipp. 2018. Virtual O&M: A Far North Queensland Innovation. *International Journal of Orientation & Mobility* 9, 1 (2018), 1–4.

[7] Virginia Braun and Victoria Clarke. 2019. Reflecting on Reflexive Thematic Analysis. *Qualitative Research in Sport, Exercise and Health* 11, 4 (2019), 589–597.

[8] John Brooke. 1996. SUS: A Quick and Dirty Usability Scale. In *Usability Evaluation in Industry*. CRC Press, 189–194.

[9] James C. Byers, A. C. Bittner, and Susan G. Hill. 1989. Traditional and Raw Task Load Index (TLX) Correlations: Are Paired Comparisons Necessary. In *Advances in Industrial Ergonomics and Safety*. Elsevier, Amsterdam, The Netherlands, 481–485.

[10] Junlong Chen, Rosella P. Galindo Esparza, Vanja Garaj, Per Ola Kristensson, and John Dudley. 2025. EnVisionVR: A Scene Interpretation Tool for Visual Accessibility in Virtual Reality.

[11] Elizabeth R Chrastil and William H Warren. 2015. Active and Passive Spatial Learning in Human Navigation: Acquisition of Graph Knowledge. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 41, 4 (2015), 1162–1178.

[12] SeungA Chung, Kyungyeon Lee, and Uran Oh. 2021. Understanding the Two-Step Nonvisual Omnidirectional Guidance for Target Acquisition in 3D Spaces. In *ISMAR*. IEEE, Piscataway, NJ, USA, 339–346.

[13] Erin Connors, Elizabeth Chrastil, Jaime Sánchez, and Lotfi Merabet. 2014. Virtual Environments for the Transfer of Navigation Skills in the Blind: a Comparison of Directed Instruction Vs. Video Game Based Learning Approaches. *Frontiers in Human Neuroscience* 8 (2014), 223.

[14] Patricia A. de Oliveira, Erich P. Lotto, Ana Grasielle D. Correa, Luis G. G. Taboada, Laisa C. P. Costa, and Roseli D. Lopes. 2015. Virtual Stage: An Immersive Musical Game for People with Visual Impairment. In *SBGAMES*. IEEE, Piscataway, NJ, USA, 135–141.

[15] Alessandro Diamanti. 2014. *Low Poly City*. https://sketchfab.com/3d-models/low-poly-city-41697300a4c643d089784b8688b2ed2c CC BY License.

[16] Agebson Rocha Façanha, Ticianne Darin, Windson Viana, and Jaime Sánchez. 2020. O&M Indoor Virtual Environments for People Who Are Blind: A Systematic Literature Review. *TACCESS* 13, 2 (2020), 1–42.

[17] Jacqui Fashimpaur, Tovi Grossman, Ben Lafreniere, Naveen Sendhilnathan, Kashyap Todi, Tianyi Wang, Ting Zhang, and Tanya R. Jonker. 2025. Squiggle: Multimodal Lasso Selection in the Real World. In *UIST*. ACM, New York, NY, USA.

[18] Rachel L. Franz, Jinghan Yu, and Jacob O. Wobbrock. 2023. Comparing Locomotion Techniques in Virtual Reality for People with Upper-Body Motor Impairments. In *ASSETS*. ACM, New York, NY, USA, 1–15.

[19] Nicholas A. Giudice. 2018. Navigating Without Vision: Principles of Blind Spatial Cognition. In *Handbook of Behavioral and Cognitive Geography*. Edward Elgar Publishing, 260–288.

[20] David Gonçalves, Manuel Piçarra, Pedro Pais, João Guerreiro, and André Rodrigues. 2023. "My Zelda Cane": Strategies Used by Blind Players to Play Visual-Centric Digital Games. In *CHI*. ACM, New York, NY, USA.

[21] João Guerreiro, Yujin Kim, Rodrigo Nogueira, SeungA Chung, André Rodrigues, and Uran Oh. 2023. The Design Space of the Auditory Representation of Objects and Their Behaviours in Virtual Reality for Blind People. *TVCG* 29, 5 (2023), 2763–2773.

[22] João Guerreiro, Daisuke Sato, Dragan Ahmetovic, Eshed Ohn-Bar, Kris M. Kitani, and Chieko Asakawa. 2020. Virtual Navigation for Blind People: Transferring Route Knowledge to the Real-World. *International Journal of Human-Computer Studies* 135 (2020), 102369.

[23] João Guerreiro, Dragan Ahmetovic, Kris M. Kitani, and Chieko Asakawa. 2017. Virtual Navigation for Blind People: Building Sequential Representations of the Real-World. In *ASSETS*. ACM, New York, NY, USA, 280–289.

[24] Sangsun Han, Pilhyoun Yoon, Miyeon Ha, and Kibum Kim. 2022. VR Wayfinding Training for People with Visual Impairment Using VR Treadmill and VR Tracker. In *VRW*. IEEE, Piscataway, NJ, USA, 596–597.

[25] Zebin Han, Xudong Wang, Baichen Liu, Qi Lyu, Zhenduo Shang, Jiahua Dong, Lianqing Liu, and Zhi Han. 2026. SeqWalker: Sequential-Horizon Vision-and-Language Navigation with Hierarchical Planning.

[26] Leona Holloway, Kim Marriott, and Matthew Butler. 2018. Accessible Maps for the Blind: Comparing 3D Printed Models with Tactile Graphics. In *CHI*. ACM, New York, NY, USA, 1–13.

[27] Bowen Huang, Yanwei Zheng, Chuanlin Lan, Dongchen Sui, Xinpeng Zhao, Xiao Zhang, Mengbai Xiao, and Dongxiao Yu. 2025. Action-Aware Visual-Textual Alignment for Long-Instruction Vision-and-Language Navigation. *TOMM* 21, 9 (2025), 1–22.

[28] Ying Ying Huang. 2010. Exploration in 3D Virtual Worlds with Haptic-Audio Support for Nonvisual Spatial Recognition. In *Human-Computer Interaction*. Springer, Brisbane, Australia, 269–272.

[29] Gabriel Ilharco, Vihan Jain, Alexander Ku, Eugene Ie, and Jason Baldridge. 2019. General Evaluation for Instruction Conditioned Navigation Using Dynamic Time Warping.

[30] Gaurav Jain, Yuanyang Teng, Dong Heon Cho, Yunhao Xing, Maryam Aziz, and Brian A. Smith. 2023. "I Want to Figure Things Out": Supporting Exploration in Navigation for People with Visual Impairments. *PACMHCI* 7, CSCW1 (2023).

[31] Puneet Jain and Christian Bayerlein. 2026. Hand to Mouth: Shifting the Bare-Minimum Accessibility Paradigm in XR Through Crip-Hacking and Crip-Aesthetics. *Ethics and Society* 20, 2 (2026), 16.

[32] Rie Kamikubo, Hernisa Kacorri, and Chieko Asakawa. 2024. "We Are at the Mercy of Others' Opinion": Supporting Blind People in Recreational Window Shopping with AI-infused Technology. In *W4A*. ACM, New York, NY, USA, 45–58.

[33] Daniel Killough, Justin Feng, Zheng Xue Ching, Daniel Wang, Rithvik Dyava, Yapeng Tian, and Yuhang Zhao. 2025. VRSight: An AI-Driven Scene Description System to Improve Virtual Reality Accessibility for Blind People. In *UIST*. ACM, New York, NY, USA.

[34] Julian Kreimeier and Timo Götzelmann. 2020. Two Decades of Touchable and Walkable Virtual Reality for Blind and Visually Impaired People: A High-level Taxonomy. *Multimodal Technologies and Interaction* 4, 4 (2020), 79.

[35] Julian Kreimeier, Pascal Karg, and Timo Götzelmann. 2020. BlindWalkVR: Formative Insights Into Blind and Visually Impaired People's VR Locomotion Using Commercially Available Approaches. In *PETRA*. ACM, New York, NY, USA, 1–8.

[36] Masaya Kubota, Masaki Kuribayashi, Renato Alexandre Ribeiro, and Shigeo Morishima. 2025. Investigating Blind People's Route Perception of Complex Pre-Mapped Routes in Virtual Reality. In *UIST*. ACM, New York, NY, USA, 1–3.

[37] Masaki Kuribayashi, Tatsuya Ishihara, Daisuke Sato, Jayakorn Vongkulbhisal, Karnik Ram, Seita Kayukawa, Hironobu Takagi, Shigeo Morishima, and Chieko Asakawa. 2023. PathFinder: Designing a Map-less Navigation System for Blind People in Unfamiliar Buildings. In *CHI*. ACM, New York, NY, USA.

[38] Masaki Kuribayashi, Seita Kayukawa, Jayakorn Vongkulbhisal, Chieko Asakawa, Daisuke Sato, Hironobu Takagi, and Shigeo Morishima. 2022. Corridor-Walker: Mobile Indoor Walking Assistance for Blind People to Avoid Obstacles and Recognize Intersections. *PACMHCI* 6, MHCI (2022).

[39] Lahav, Gedalevitz, Battersby, Brown, Evett, and Merritt. 2013. Pre-planning Navigation with Virtual Environments for People Who Are Blind Using Wii Technology. In *ICVR*. IEEE, Piscataway, NJ, USA, 229–234.

[40] Orly Lahav. 2022. Virtual Reality Systems as an Orientation Aid for People Who Are Blind to Acquire New Spatial Information. *Sensors* 22, 4 (2022).

[41] Diogo Lança, Manuel Piçarra, Inês Gonçalves, Uran Oh, André Rodrigues, and João Guerreiro. 2024. Speed-of-Light VR for Blind People: Conveying the Location of Arm-Reach Targets. In *ASSETS*. ACM, New York, NY, USA.

[42] Samuel Martin, Pooyan Fazli, and Hasti Seifi. 2026. VisionPulse: A Virtual Reality System Enabling Accessible Discovery and Navigation for Blind and Low Vision Users.

[43] Luca-Maxim Meinhardt, Lina Madlin Weilke, Maryam Elhaidary, Julia von Abel, Paul D. S. Fink, Michael Rietzler, Mark Colley, and Enrico Rukzio. 2025. Light My Way: Developing and Exploring a Multimodal Interface to Assist People With Visual Impairments to Exit Highly Automated Vehicles. In *CHI*. ACM, New York, NY, USA.

[44] Yusuke Miura, Erwin Wu, Masaki Kuribayashi, Hideki Koike, and Shigeo Morishima. 2023. Exploration of Sonification Feedback for People with Visual Impairment to Use Ski Simulator. In *AHs*. ACM, New York, NY, USA, 147–158.

[45] Daniel R. Montello. 1998. A New Framework for Understanding the Acquisition of Spatial Knowledge in Large-Scale Environments. In *Spatial And Temporal Reasoning In Geographic Information Systems*. Oxford University Press.

[46] Karin Müller, Christin Engel, Claudia Loitsch, Rainer Stiefelhagen, and Gerhard Weber. 2022. Traveling More Independently: A Study on the Diverse Needs and Challenges of People with Visual or Mobility Impairments in Unfamiliar Indoor Environments. *TACCESS* 15, 2 (2022).

[47] Ruth G Nagassa, Matthew Butler, Leona Holloway, Cagatay Goncu, and Kim Marriott. 2023. 3d Building Plans: Supporting Navigation by People Who Are Blind or Have Low Vision in Multi-storey Buildings. In *CHI*. ACM, New York,

NY, USA, 1–19.
[48] Vishnu Nair, Jay L Karp, Samuel Silverman, Mohar Kalra, Hollis Lehv, Faizan Jamil, and Brian A. Smith. 2021. NavStick: Making Video Games Blind-Accessible Via the Ability to Look Around. In *UIST*. ACM, New York, NY, USA, 538–551.
[49] Vishnu Nair, Shao-en Ma, Ricardo E. Gonzalez Penuela, Yicheng He, Karen Lin, Mason Hayes, Hannah Huddleston, Matthew Donnelly, and Brian A. Smith. 2022. Uncovering Visually Impaired Gamers' Preferences for Spatial Awareness Tools Within Video Games. In *ASSETS*. ACM, New York, NY, USA, 1–16.
[50] Ruowen Niu, Ruishen Zheng, Chen Liang, and Minghui Liu. 2025. Exploring Joint Effects of Locomotion Continuity and Wayfinding Assistance in Non-Embodied VR Game Navigation. In *CHI*. ACM, New York, NY, USA.
[51] OpenAI. 2024. Hello GPT-4o. https://openai.com/index/hello-gpt-4o/.
[52] Konstantinos Papadopoulos, Eleni Koustriava, and Panagiotis Koukourikos. 2018. Orientation and Mobility Aids for Individuals with Blindness: Verbal Description Vs. Audio-tactile Map. *Assistive Technology* 30, 4 (2018), 191–200.
[53] Romedi Passini and Guyltne Proulx. 1988. Wayfinding Without Vision. *Environment and Behavior* 20, 2 (1988), 227–252.
[54] Manuel Piçarra, André Rodrigues, and João Guerreiro. 2023. Evaluating Accessible Navigation for Blind People in Virtual Environments. In *CHI'EA*. ACM, New York, NY, USA.
[55] Lorenzo Picinali, Amandine Afonso, Michel Denis, and Brian F.G. Katz. 2014. Exploration of Architectural Spaces by Blind People Using Auditory Virtual Reality for the Construction of Spatial Knowledge. *International Journal of Human-Computer Studies* 72, 4 (2014), 393–407.
[56] Navid Rajabi and Jana Kosecka. 2025. TRAVEL: Training-Free Retrieval and Alignment for Vision-and-Language Navigation.
[57] Niyati Rawal, Roberto Bigazzi, Lorenzo Baraldi, and Rita Cucchiara. 2024. UNMuTe: Unifying Navigation and Multimodal Dialogue-like Text Generation.
[58] Renato Alexandre Ribeiro, Inês Gonçalves, Manuel Piçarra, Letícia Seixas Pereira, Carlos Duarte, André Rodrigues, and João Guerreiro. 2024. Investigating Virtual Reality Locomotion Techniques with Blind People. In *CHI*. ACM, New York, NY, USA.
[59] Roy A Ruddle, Stephen J Payne, and Dylan M Jones. 1997. Navigating Buildings in "desk-top" Virtual Environments: Experimental Investigations Using Extended Navigational Experience. *Journal of Experimental Psychology: Applied* 3, 2 (1997), 143–159.
[60] JAIME SÁNCHEZ and MAURICIO LUMBRERAS. 1999. Virtual Environment Interaction Through 3D Audio by Blind Children. *CyberPsychology & Behavior* 2, 2 (1999), 101–111.
[61] Daisuke Sato, Uran Oh, João Guerreiro, Dragan Ahmetovic, Kakuya Naito, Hironobu Takagi, Kris M Kitani, and Chieko Asakawa. 2019. NavCog3 in the Wild: Large-scale Blind Indoor Navigation Assistant with Semantic Features. *TACCESS* 12, 3 (2019), 14.
[62] David W. Schloerb, Orly Lahav, Joseph G. Desloge, and Mandayam A. Srinivasan. 2010. BlindAid: Virtual Environment System for Self-reliant Trip Planning and Orientation and Mobility Training. In *HAPTIC*. IEEE, Piscataway, NJ, USA, 363–370.
[63] Yoshikazu Seki and Tetsuji Sato. 2011. A Training System of Orientation and Mobility for Blind People Using Acoustic Virtual Reality. *TNSRE* 19, 1 (2011), 95–104.
[64] Alexa F. Siu, Mike Sinclair, Robert Kovacs, Eyal Ofek, Christian Holz, and Edward Cutrell. 2020. Virtual Reality Without Vision: A Haptic and Auditory White Cane to Navigate Complex Virtual Worlds. In *CHI*. ACM, New York, NY, USA, 1–13.
[65] Maarten Someren, Yvonne Barnard, and Jacobijn Sandberg. 1994. *The Think Aloud Method - A Practical Guide to Modelling Cognitive Processes*. Academic Press, San Diego, CA, USA.
[66] Yanjun Sun, Yue Qiu, and Yoshimitsu Aoki. 2025. DynamicVLN: Incorporating Dynamics Into Vision-and-Language Navigation Scenarios. *Sensors* 25, 2 (2025).
[67] Unity Technologies. 2025. Physics.Raycast. https://docs.unity3d.com/ScriptReference/Physics.Raycast.html.
[68] Malgorzata Telesinska. 2023. Multimodal 3D Printed Urban Maps for Blind People. Evaluations and Scientific Investigations. In *ASSETS*. ACM, New York, NY, USA, 1–7.
[69] Lauren Thevin, Carine Briant, and Anke M. Brock. 2020. X-Road: Virtual Reality Glasses for Orientation and Mobility Training of People with Visual Impairments. *TACCESS* 13, 2 (2020).
[70] Chanh Minh Tran, Nguyen Gia Bach, Phan Xuan Tan, Eiji Kamioka, and Manami Kanamaru. 2024. Enabling Self-Practice of Digital Audio–Tactile Maps for Visually Impaired People by Large Language Models. *Electronics* 13, 12 (2024), 2395.
[71] Xiyue Wang, Seita Kayukawa, Hironobu Takagi, and Chieko Asakawa. 2022. BentoMuseum: 3D and Layered Interactive Museum Map for Blind Visitors. In *ASSETS*. ACM, New York, NY, USA, 1–14.
[72] Yuhang Zhao, Cynthia L. Bennett, Hrvoje Benko, Edward Cutrell, Christian Holz, Meredith Ringel Morris, and Mike Sinclair. 2018. Enabling People with Visual Impairments to Navigate Virtual Reality with a Haptic and Auditory Cane Simulation. In *CHI*. ACM, New York, NY, USA, 1–14.
[73] Yuhang Zhao, Edward Cutrell, Christian Holz, Meredith Ringel Morris, Eyal Ofek, and Andrew D. Wilson. 2019. SeeingVR: A Set of Tools to Make Virtual Reality More Accessible to People with Low Vision. In *CHI*. ACM, New York, NY, USA, 1–14.
[74] Yuhang Zhao, Sarit Szpiro, and Shiri Azenkot. 2015. ForeSee: A Customizable Head-Mounted Vision Enhancement System for People with Low Vision. In *ASSETS*. ACM, New York, NY, USA, 239–249.